Assembling the IN-17 sockets

I don't want to give advice to anyone about soldering nixie tubes, in this document I just show how it works for me.

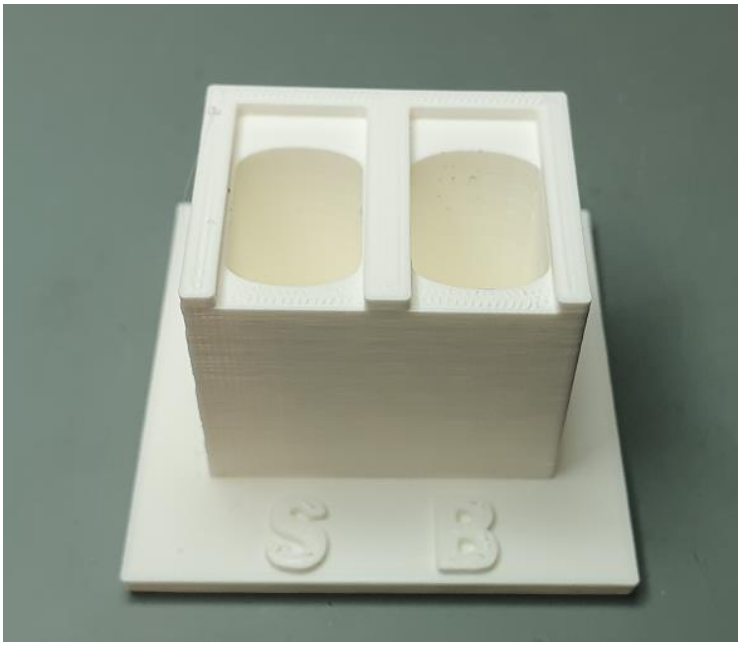
Nixies are expensive these days and are the most valuable parts of this project, so avoiding mistakes is important.

WARNING: The dimensions of tubes of the same type can vary considerably. Don't use nixies wider than 13mm for this project if you cannot compensate the difference with the adjacent tubes!

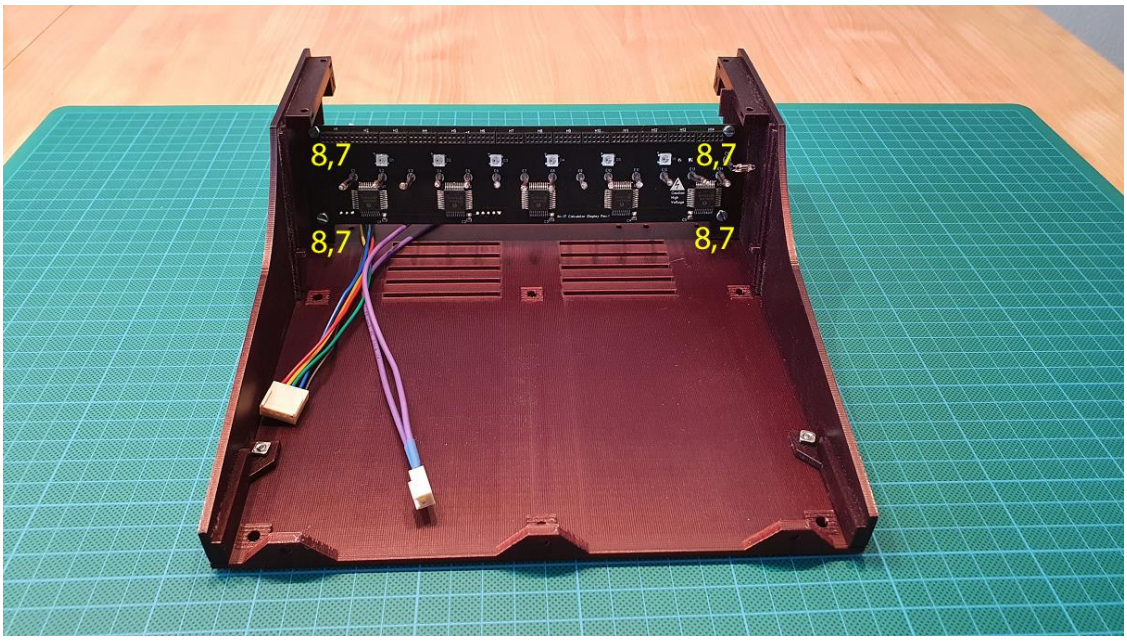
I didn't have any problems with any of my (non-matched) tubes.

The following steps worked for me:

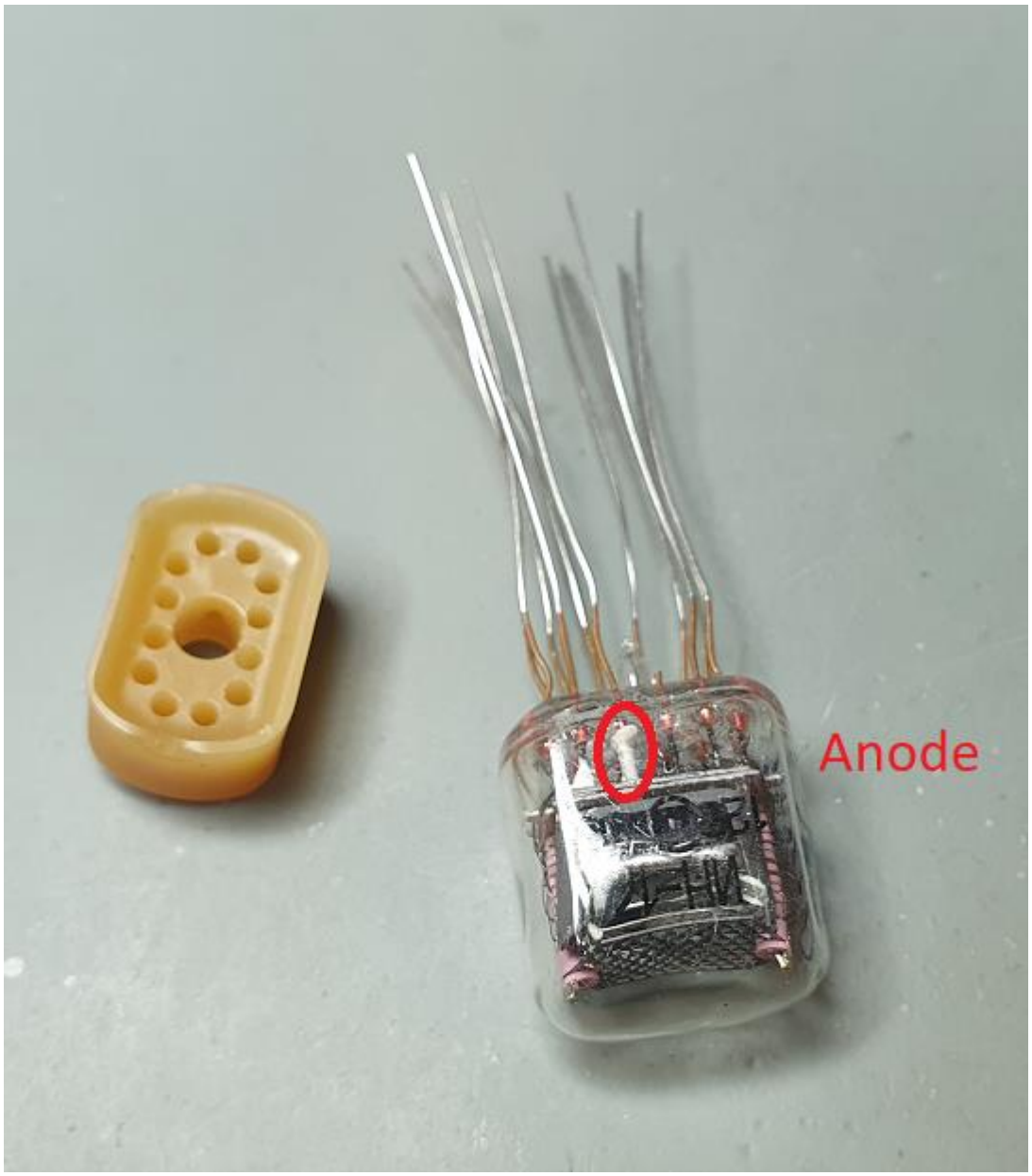
1. Print the IN-17 tool



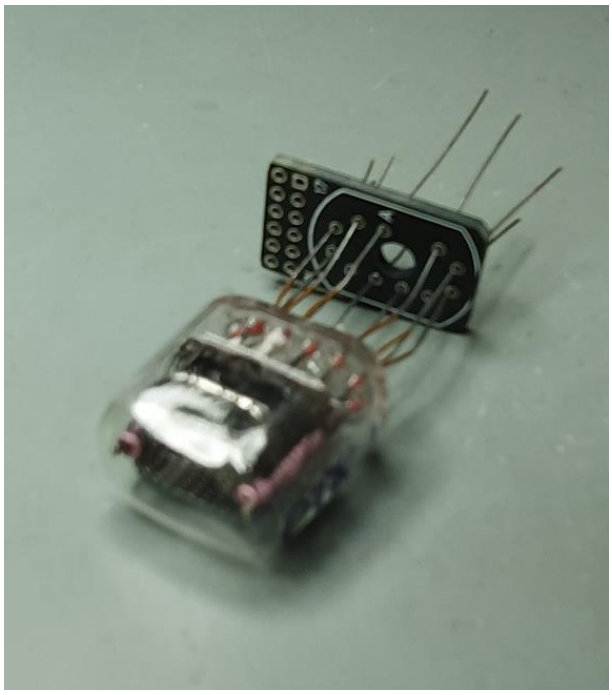
2. Start assembling the calculator and stop **after** the step “Install the driver and display board” (see Docs/Assembling the IN-17 calculator.pdf) This is important because we can check if the tubes fit into the case.



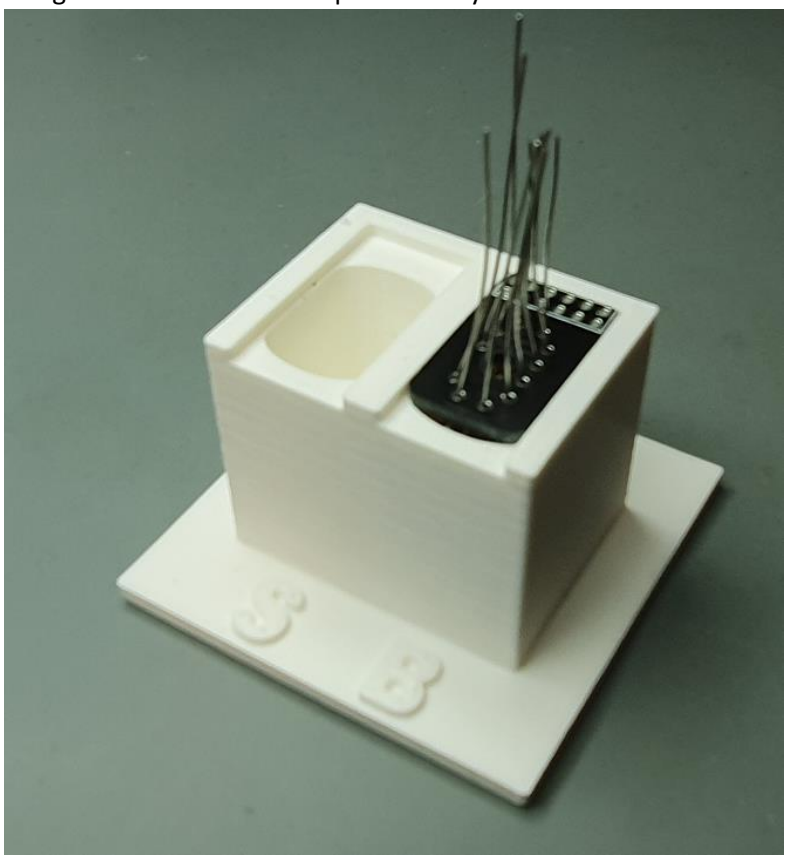
3. Carefully remove the plastic spacer and find the anode lead (**double-check with IN-17 datasheet**!) Leave the anode lead as it is, take the next (adjacent) lead and cut about 1mm from the free end of the lead. Then take the next lead and cut 2mm, take the next lead and cut 3mm and so on…



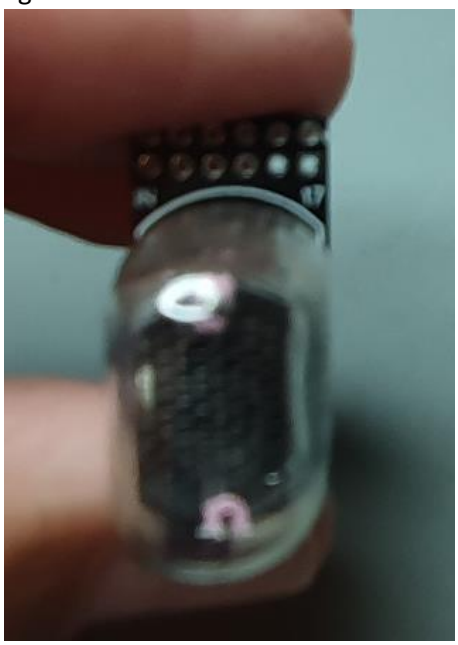
4. Introduce the leads in a socket board starting with the anode lead. On the board the anode pad is marked with an A

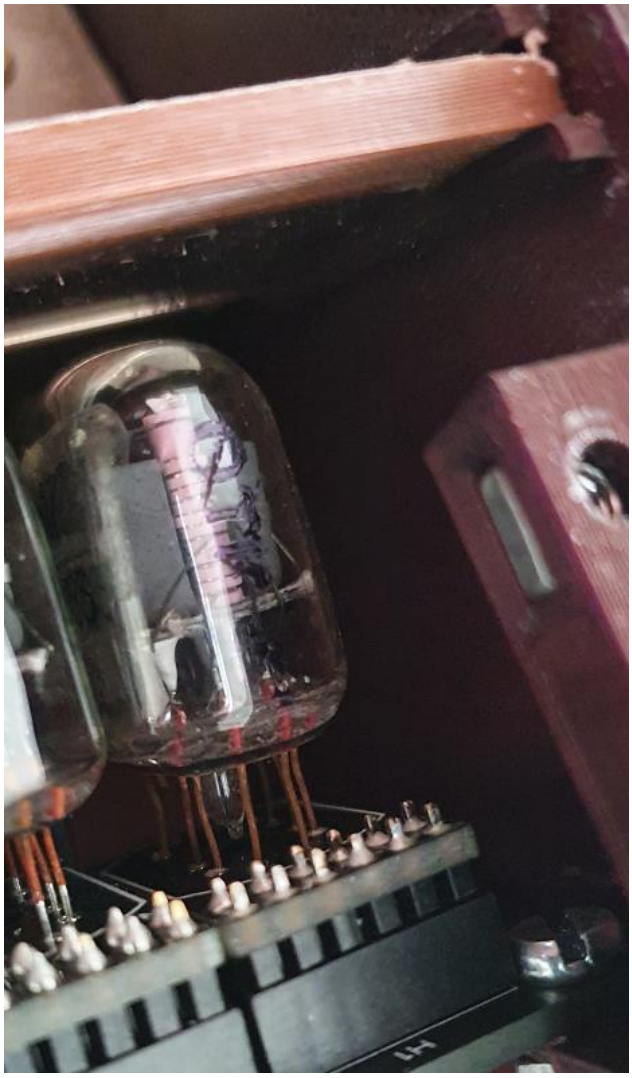


5. Bring the socket board into position. Try first slot **S**. If the nixie does not fit, try slot **B**



6. **Push the tube all the way down**. Unfortunately, the tool does only help for setting the distance between socket board and tube but not for the alignment. I start soldering only one lead and then I check the alignment. If the alignment is ok, I solder the opposite lead, check again and so on.



7. Solder the header

8. Verify that the nixie with the socket fits into the case by installing the nixie and the front panel.